Table 3: Turnover Regressions with Standardized Disagreement

	Baseline	Rec × DI	Rec × DI	Early Rec.	Early Rec.
	(1)	(2)	(3)	(4)	(5)
Disagreement (z-score)	0.075	0.054	0.054	0.069	0.069
	(0.047)	(0.039)	(0.039)	(0.047)	(0.047)
Recession		0.051			
		(0.048)			
Disagreement × Recession		0.176***	0.188***		
		(0.058)	(0.061)		
Early Rec. (first 2 qtrs)				0.014	
				(0.035)	
Disagreement × Early Rec.				0.095*	0.098*
				(0.051)	(0.051)
Marginal effect of DI in recession		0.230	0.242	0.164	0.167
Constant	0.311***	0.301***	0.305***	0.309***	0.309***
	(0.023)	(0.022)	(0.020)	(0.023)	(0.022)
Observations	158	158	158	158	158
R^2	0.212	0.352	0.344	0.235	0.235

Sample period: 1982Q2–2021Q3. Disagreement is standardized within sample. Newey–West (lag 4) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

most strongly to disagreement at the onset of downturns.

The results are robust to controlling for aggregate market volatility (VIX) and market returns; these specifications are reported in Appendix O1.4.

Overall, the evidence supports the model's central implication: while trading volume increases with belief dispersion, this sensitivity is substantially stronger in recessions, particularly around state transitions.

6 Quantitative analysis

We now consider the quantitative implications. We extend the baseline formulation in Section 3: First, productivity growth x_t now follows a more general process. This is necessary to capture the empirical patterns of dividend growth. Second, we introduce a force that induces a non-degenerate stationary wealth distribution. A force like this is needed so that belief heterogeneity survives in the ergodic distribution of the model. We do so

through Uzawa endogenous discounting for convenience. This specific approach is not indispensable; we can obtain non-degeneracy through birth-death processes for example.

Belief process with a continuum of states. We start by specifying a continuous Markov process for x_t under both objective and subjective beliefs. Let $\hat{x}_t \equiv \log x_t$ denote log aggregate productivity growth. Under the objective measure, $\hat{x}_t$ follows:

$$\hat{x}_{t+1} = \mu_x + \sigma_x \epsilon_{x,t+1}. \tag{40}$$

where $\epsilon_{x,t+1} \overset{\text{iid}}{\sim} N(0,1)$. Thus, productivity growth is iid under the objective measure.

In turn, subjective beliefs about $\hat{x}_t$ for household $i \in \mathcal{I}$ are given by:

$$\hat{x}_{t+1} = \mu_{x,i} + \rho_{x,i}(\hat{x}_t - \mu_{x,i}) + v_{i,t} + \sigma_{x,i}\epsilon_{x,i,t+1}, \tag{41}$$

where $\epsilon_{x,i,t+1} \overset{\text{iid}}{\sim} N(0,1)$. Beliefs deviate from the process (40) in two critical ways: First, the subjective conditional expectation responds to current productivity growth $\hat{x}_t$. This dimension is important to allow households to over- or under-react to past information, consistent with Fact 2 in Section 2. Overreaction is controlled by $\rho_{x,i}$. For instance, a positive $\rho_{x,i}$ implies the household overreacts to productivity news, given our assumption that productivity growth is iid under the objective measure.

Second, beliefs are also exposed to a persistent *sentiment shock* v_t, which evolve according to

$$v_{i,t+1} = \rho_{v,i} v_{i,t} + \sigma_{v,i}\epsilon_{v,i,t+1} \tag{42}$$

where $\epsilon_{v,i,t+1} \overset{\text{iid}}{\sim} N(0,1)$, and $\epsilon_{v,i,t}$ and $\epsilon_{x,i,t}$ are uncorrelated.

The sentiment is independent of TFP growth. Thus, this shock leads to fluctuations in the degree of optimism unrelated to fundamentals. Introducing this shock is necessary to quantitatively match the volatility of subjective TFP growth expectations relative to the objective one—Fact 1 in Section 2. While extrapolation, through higher values of $\rho_{x,i}$, increases the volatility of subjective expectations, it is insufficient, so the sentiment is needed.

We note that the belief system in this section is a hybrid of the one in our taxonomy. It encompasses both extrapolation and time-dependent optimism through sentiment shock. We let the survey data discipline the strength of each force.

Endogenous discounting. To ensure the wealth distribution is stationary, we assume that households' subjective discount rate responds to their consumption share:

$$\beta_{i,t} = \beta e^{\kappa\left(1-\frac{\mu_i \tilde{C}_{i,t}}{\tilde{Y}_t}\right)}. \tag{43}$$

This assumption, a form of Uzawa (2017) preferences, implies that households' marginal propensity to consume increases with their share of consumption, ensuring that no investor type concentrates all the wealth asymptotically. Following Schmitt-Grohé and Uribe (2003), we assume that $\beta_{i,t}$ depends on the average consumption share of type-i households, so households take the process for $\beta_{i,t}$ as given.[19]

Model solution. We describe the model characterization with a continuum of states and Uzawa preferences in Appendix O2. Previous results are essentially unchanged. As in the binary case, an exact closed-form solution is unavailable away from log preferences. We compute the solution using a third-order perturbation around the non-stochastic steady state. A third-order perturbation is indispensable to capture time variation in expected returns.

Calibration. We use the following calibration, where parameters are expressed in quarterly terms. Preferences and technology parameters are standard. *Preferences*: We set $\beta = 0.978$ to match an unconditional annualized risk-free rate of 1%. The risk aversion is set to $\gamma = 10.0$ and the EIS to $\psi = 2.0$, typical values in the macro-finance literature. We choose the labor disutility parameter ξ to normalize the average hours to 1 and set the Frish elasticity to one, $\nu = 1$, a standard value in the literature. *Technology:* We set $\alpha = 0.66$ and choose μ_x and σ_x to match the average and standard deviation of annual consumption growth of 2.1% and 2.7%, respectively.

The calibration of the belief processes is novel. *Beliefs:* We focus on the case $I = 2$, and assume that households of type $i = 2$ have rational beliefs, which allows for some rational agents in the environment. We set $\mu_{x,1} = \mu_x$ and $\sigma_{x,1} = \sigma_x$, so households agree on the mean and conditional volatility of productivity growth. We set $\mu_2 = 0.1$, so the fraction of rational agents in the population is 10%, consistent with the fact that average beliefs in surveys of households and analysts show substantial deviations from rational expectations. We set $\rho_{v,1}$ and $\sigma_{v,1}$ to match the persistence of subjective dividend growth and the share of variance explained by movements in expectations, based on estimations of

[19]This mechanism is often used in small open-economy models to ensure a stationary distribution of external debt.

De La O and Myers (2021). In turn, we set $\rho_{x,1}$ to match the correlation between subjective expectations and current productivity growth observed in the data. This calibration is agnostic about the relevant belief system: the subjective belief moments pin down these parameters, determining whether non-rational beliefs are extrapolation or rank-preserving. Finally, we set $\kappa = 0.5\%$, consistent with the mean-reversion in consumption shares in Gârleanu and Panageas (2015).

6.1 Biases in productivity growth expectations

In the model, investors hold distorted beliefs about productivity growth. These distortions propagate to other variables — such as dividends and returns — through the likelihood-ratio representation:

$$\mathbb{E}_{i,t}[Z_{t+1}] = \mathbb{E}_t[\ell_{i,t+1} Z_{t+1}], \tag{44}$$

where $\ell_{i,t+1}$ is the likelihood-ratio between subjective and objective beliefs. Hence, a single distortion in productivity growth beliefs induces distortions in expectations of all equilibrium objects.[20]

It is therefore important to verify that the model is consistent with the empirical properties of subjective productivity growth expectations.

Coibion–Gorodnichenko regressions. We assess this using Coibion–Gorodnichenko (CG) regressions (Coibion and Gorodnichenko, 2015), which test for systematic forecast biases by regressing forecast errors on forecast revisions:

$$\Delta prod_{t+h} - \mathbb{E}_{i,t}[\Delta prod_{t+h}] = c + \beta_{CG} \left(\mathbb{E}_{i,t}[\Delta prod_{t+h}] - \mathbb{E}_{i,t-1}[\Delta prod_{t+h}] \right) + u_{t+h}. \tag{45}$$

Under full-information rational expectations, $\beta_{CG} = 0$. A negative estimate indicates over-extrapolation: agents revise forecasts too strongly in the direction of recent news.

Empirical implementation. As direct survey data on TFP growth expectations are limited, we consider two alternative measures of productivity growth using quarterly data

[20] Cui, De La O and Myers (2024) estimate an empirical counterpart to the likelihood ratio and show that it explains a large fraction of cross-sectional variation in survey expectations.

Table 4: Coibion–Gorodnichenko Regressions for Productivity Measures

	Labor Productivity	TFP Proxy
Forecast revision coefficient β_{CG}	-0.63***	-0.50**
	(0.030)	(0.198)
Constant	-0.15	-0.25
	(0.289)	(0.259)
Observations	86	86
R^2	0.3697	0.0579
Adj. R^2	0.3622	0.0467

Notes: Each column reports OLS estimates of $\text{FE}_t = c + \beta\,\text{Revision}_t + u_t$, where $h = 1$ quarter. Reported standard errors (in parentheses) are Newey–West HAC with lag 4. Labor productivity uses $\Delta y - \Delta n$. TFP proxy uses $\Delta y - \alpha\Delta n$ with $\alpha = 2/3$. Sample: 2004Q1–2025Q2. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (based on HAC p-values).

from the Survey of Professional Forecasters (SPF):

$$\Delta prod_{t+1}^{LP} = \Delta\hat{y}_{t+1} - \Delta\hat{n}_{t+1}, \tag{46}$$

$$\Delta prod_{t+1}^{TFP} = \Delta\hat{y}_{t+1} - \alpha\,\Delta\hat{n}_{t+1}, \tag{47}$$

where $\Delta\hat{y}$ is GDP growth, $\Delta\hat{n}$ is employment growth, and we set $\alpha = 2/3$.

The first measure corresponds to labor productivity growth, and the second is a TFP proxy consistent with how TFP growth is defined in the model. Table 4 reports the results. For labor productivity, we estimate $\hat{\beta}_{CG} = -0.63$, and for the TFP proxy, $\hat{\beta}_{CG} = -0.50$, both statistically significant.

Model comparison. We perform the same CG regression using simulated data from the calibrated model. The model implies

$$\hat{\beta}_{CG}^{\text{model}} = -0.695.$$

This value is very close to the empirical estimate for labor productivity and lies within one standard error of the estimate for the TFP proxy.

These results support the model's mechanism. The same distortions that drive endogenous movements in labor demand and risk premia also generate empirically realistic forecast biases in productivity growth.

Table 5: Unconditional moments

	Rational Beliefs				Subjective Beliefs							
	Endowment		Production		Homogeneous		Heterogeneous		High elast.		Data (± 2 s.e.)	
Variables	**Mean**	**Std**	**Mean**	**Std**	**Mean**	**Std**	**Mean**	**Std**	**Mean**	**Std**	**Mean**	**Std**
Interest rate	10.2	0.0	10.0	0.7	10.1	1.3	1.7	6.1	1.6	6.1	[0.6, 3.0]	[3.7, 6.0]
Excess returns (equity)	0.7	2.7	1.2	7.2	1.2	8.3	5.2	15.8	5.3	15.6	[3.5, 9.3]	[15.6, 21.4]
Consumption growth	2.1	2.7	2.1	2.7	2.1	2.8	2.1	2.8	2.1	3.0	[1.2, 3.1]	[2.0, 3.4]
Dividend growth	2.1	2.7	2.1	9.3	2.1	10.0	2.1	10.0	2.1	9.8	[-0.3, 3.4]	[6.9, 11.2]
Log hours	0.0	0.0	0.2	0.0	0.2	1.3	0.2	1.2	0.6	1.9	–	[1.7, 2.7]

Note: The table reports annualized unconditional moments in percentage points. Column 1 is the endowment benchmark with rational beliefs; Column 2 adds production with labor chosen in advance under rational beliefs; Column 3 is the homogeneous-beliefs version of the production model; Column 4 is the heterogeneous-beliefs benchmark calibration; Column 5 is the high-labor-elasticity robustness. The data columns report point estimate ± 2 bootstrap standard errors. Detailed data construction, sample windows, and bootstrap methodology are provided in the Data Appendix (Appendix O1.6).

Overall, even though the model is not directly calibrated to productivity-expectations moments, it reproduces the key bias pattern observed in the data.

6.2 Unconditional moments

Next, we study the model's ability to match unconditional moments. To isolate the role of each ingredient, we start from a stripped-down version of the model and progressively add features until we reach the complete model. This progression allows us to discuss the role of each ingredient in the model. Table 5 presents the results under each formulation. The table also reports ± 2 standard-error intervals for the data moments.

Column 1 presents the stripped-down version of the model with a single representative rational investor, for which hiring is done after TFP is known and labor supply is inelastic. This economy behaves as an Epstein-Zin version of the endowment economy studied by Mehra and Prescott (1985). The model fails to generate a sizeable equity premium and excess stock volatility. Moreover, without the hiring friction, the volatility of consumption and dividends coincide. Given that rational beliefs are iid, the price-dividend ratio is constant, so return volatility equals dividend volatility.

Column 2 introduces the timing of hires assumption and the elastic labor supply. The operating leverage channel endogenously generates differences in volatility between consumption and dividends in line with the data, even though this is not a targeted moment. With iid beliefs, there is no time variation in the price of risk, so the model fails to gen-

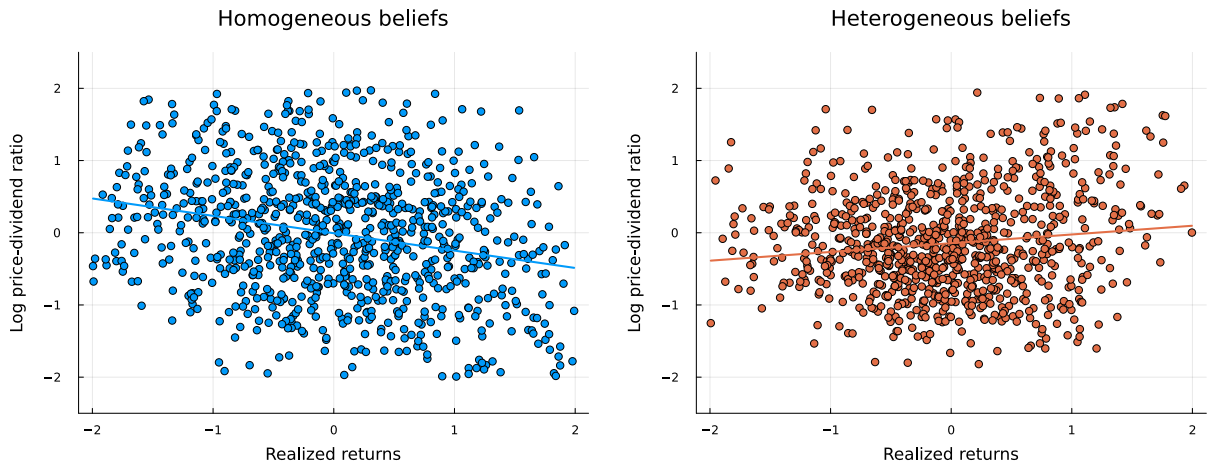

Figure 7: Price amplification: homogeneous vs. heterogeneous beliefs

Note: Figure shows a scatterplot of the log price-dividends and realized log returns in a 1,000-period simulation of the model. The left panel shows the simulation for the model with homogeneous beliefs, and the right panel shows the simulation for the model with heterogeneous beliefs. We standardized both variables and removed outliers to improve the visualization.

erate fluctuations in hours. Moreover, the hiring friction implies that dividend growth follows a mean-reverting process.[21] Without time-varying risk premia, mean-reversion in dividends causes the price-dividend ratio to negatively correlate with realized returns. This counterfactual behavior dampens return volatility relative to dividend volatility.

Column 3 considers the role of non-rational beliefs calibrated to match the survey data, while abstracting from differences in beliefs. In this case, the model does generate movements in hours, given that subjective beliefs are volatile, consistent with the discussion in Section 4. However, deviations from rational expectations are insufficient to generate an equity premium and stock return volatility in line with the data. Movements in subjective beliefs only attenuate the negative correlation between returns and the price-dividend ratio, so return volatility is still dampened relative to dividend volatility.

Column 4 introduces heterogeneity in beliefs. It brings along rational investors and investors with distorted beliefs. The model now generates the level and volatility of asset prices consistent with the evidence, with significant volatility in hours. To obtain a high equity premium, it is crucial to generate excess volatility of returns. This requires the price-dividend ratio to have the right comovement with returns. Figure 7 shows that this is the case for the model with heterogeneous beliefs, but not for the model with homogeneous beliefs. This success is explained by the time-varying movements in the relative wealth of investors.

[21] For models with mean-reverting dividend process, see e.g. Menzly, Santos and Veronesi (2004) and Santos and Veronesi (2006).

Column 5 considers an alternative calibration with a higher labor supply elasticity, increasing it from 1.0 to 2.0, the upper limit of the estimates of the macro labor supply elasticity in Keane and Rogerson (2012). Increasing the labor supply elasticity is motivated by the presence of unmodeled frictions: it is akin to increasing the real wage rigidities. While asset-price figures do not change substantially, the volatility of hours almost doubles, aligning the volatility of hours with that of the data.

Overall, aside from a modest overprediction of risk-free-rate volatility, the model with heterogeneous beliefs matches key asset-pricing and business-cycle moments.

6.3 Excess volatility and return predictability

The model's version with heterogeneous beliefs successfully generates empirically relevant levels of excess volatility, i.e., returns that are more volatile than cash flows. Following Cochrane (1992), we can use a Campbell-Shiller approximation to decompose the variance of the price-dividend ratio into a cash-flow and a discount-rate component:

$$Var[pd_t] = Cov\left[\sum_{k=1}^{\infty} \rho^k \Delta d_{t+k}, pd_t\right] - Cov\left[\sum_{k=1}^{\infty} \rho^k r_{t+k}, pd_t\right], \tag{48}$$

where pd_t is the log price-dividend ratio, Δd_{t+1} denotes log dividend growth, and r_{t+1} denotes realized log returns. Expression (48) connects volatility and predictability. It shows that movements in the price-dividend ratio either predict changes in dividend growth or future returns. To quantify the relative importance of movements in discount rates, consider the share of variance explained by movements in expected returns, defined as $\beta_r \equiv -\frac{Cov\left[\sum_{k=1}^{\infty} \rho^k r_{t+k}, pd_t\right]}{Var[pd_t]}$. As shown, e.g., in Cochrane (2011), empirical evidence suggests that $\beta_r \approx 1$, so discount rates explain most of the price-dividend ratio variance.

Given that β_r corresponds to the slope of a regression of cumulative returns on the price-dividend ratio, the importance of discount rates to explain return volatility is tightly connected to the ability of the price-dividend ratio in predicting future long-run returns. Figure 8 shows that the model with heterogeneous beliefs can capture the predictability patterns observed in the data. The figure shows a strong negative association between the price-dividend ratio and cumulative returns in model-simulated data. Moreover, the coefficient β_r in models (4) and (5), as defined in Table 5, is given by $\beta_r = 0.86$ and $\beta_r = 0.89$, respectively. In contrast, we have $\beta_r = -0.15$ for model (2), so the price-dividend ratio predicts returns with the wrong sign, and $\beta_r = 0.26$ for model (3), which has the correct sign but movements in cash flows drive the majority of the fluctuations in the price-dividend ratio. Hence, only versions of the model with belief heterogeneity

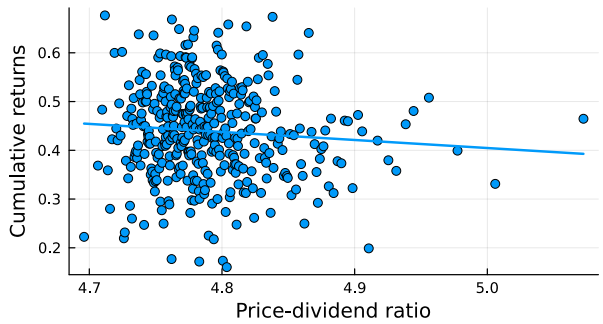

Figure 8: Return predictability

Dependent Variable:	cum_ex_returns			
Model:	(1)	(2)	(3)	(4)
Variables				
labor_income	-2.07		0.33	
	(0.94)		(1.31)	
p/d		-0.29		-0.04
		(0.12)		(0.17)
Observations	4,000	4,000	4,000	4,000

Table 6: Labor and p/d ratio regressions

Note: The left panel shows a binscatter plot based on model simulations of the price-dividend ratio and a measure of future cumulative returns, $\sum_{k=1}^{T} \rho^{k-1} r_{t+k}$, for $T = 60$ quarters. We sort the first 4,000 observations by the price-dividend ratio into 400 equal-count bins and plot bin averages. The right panel shows the regression coefficients of future cumulative excess returns for $T = 60$ on the price-dividend ratio and labor income. The data for the first two columns was simulated under the objective measure, while the data for the last two columns was simulated under subjective beliefs. Entries report Monte Carlo mean coefficients, and parentheses report the standard deviation across simulations.

generate a degree of return predictability that aligns with the data.

Labor income and return predictability. Time variation in risk premia generates fluctuations in hours worked in our setting. As a result, in addition to the price-dividend ratio, movements in labor income should also predict excess returns. Table 6 reports the results of a regression of cumulative excess returns on the price-dividend ratio and labor income in our simulated data. Column 1 shows that periods of high labor income are associated with lower future returns. This finding is consistent with the evidence from Santos and Veronesi (2006), who show that a high labor income share predicts lower aggregate stock returns. Similarly, Belo et al. (2023) document that the aggregate hiring rate of publicly traded firms negatively predicts stock market excess returns.[22] Our model links this evidence to fluctuations in subjective beliefs. As discussed in Section 4, waves of optimism lead to both an increase in labor income and a compression of risk premia, jointly generating predictability from labor income and the price-dividend ratio.

Subjective vs. objective predictability. The predictability results above are from the perspective of an econometrician. Recent work by Nagel and Xu (2023) shows that return predictability looks very different under subjective beliefs: the standard predictors of returns are only weakly associated with survey-based measures of subjective risk pre-

[22] Belo et al. (2014) and Favilukis and Lin (2016) show similar patterns for the cross-section of stock returns and U.S. states. For recent evidence using administrative data, see Meeuwis, Papanikolaou, Rothbaum and Schmidt (2024).

mia. While expected excess returns are countercyclical under the objective measure, they appear acyclical under subjective beliefs.

Table 6 shows that the model replicates these empirical patterns. Columns 1 and 2 report predictability regressions using labor income and the price-dividend ratio as predictors under the objective measure. In contrast, Columns 3 and 4 present the same regressions when the model is simulated under subjective beliefs. Consistent with Nagel and Xu (2023), standard predictors are only weakly associated with future returns under subjective beliefs. The coefficient on labor income has the wrong sign and is statistically insignificant, while the coefficient on the price-dividend ratio is near zero. Thus, our model generates subjective risk premia that are essentially acyclical, which aligns with survey evidence. As discussed in Section 4, this acyclicity arises because subjective expectations of future cash flows closely track asset price movements, dampening fluctuations in perceived risk premia.

7 Conclusion

When asked about the nature of business cycles, Thomas Sargent[23], a rational expectations pioneer, answered:[24]

> "[...] economists have been working hard to refine rational expectations theory. [...] An influential example of such work is the 1978 QJE paper by Harrison and Kreps. [...], for policymakers to know whether and how they can moderate bubbles, we need to have well-confirmed quantitative versions of such models up and running."

This response embraces the idea that "belief heterogeneity" matters but also calls for quantitative models that link beliefs with the real economy.

This paper responds to that call and adapts a standard real business cycle and asset pricing model to fit the narrative that waves of optimism and pessimism drive the business cycle. We argue that a combination of heterogeneity in beliefs, with substantial extrapolation, coupled with an operating leverage channel, can successfully reproduce business cycle, asset pricing and survey data patterns. Extrapolative beliefs and hetero-

[23]Hyman Minsky was Thomas Sargent's undergraduate advisor. Whereas Sargent departs methodologically and calls for a quantitative approach to economic research, there is an agreement in their views regarding nature of business cycles.

[24]Interview with Thomas Sargent, *The Region*, August 26, 2010. Available at https://www.minneapolisfed.org/article/2010/interview-with-thomas-sargent.

geneity deliver the amplification that is key for this success. Extrapolation adds to risk build up.

We foresee that our framework can be extended to study the interaction with other forms of amplification. Namely, nominal rigidities open the door to amplification through aggregate demand externalities. Additional financial frictions, such as margin calls or short-selling constraints, can amplify asset prices through fire-sale and pecuniary externalities that impact wealth and risk-taking capacity. We expect our conclusions to remain relevant in contexts with these additional frictions.

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